

STC89C52 Microcontroller – CPU and Architecture Study

STC89C52 Microcontroller

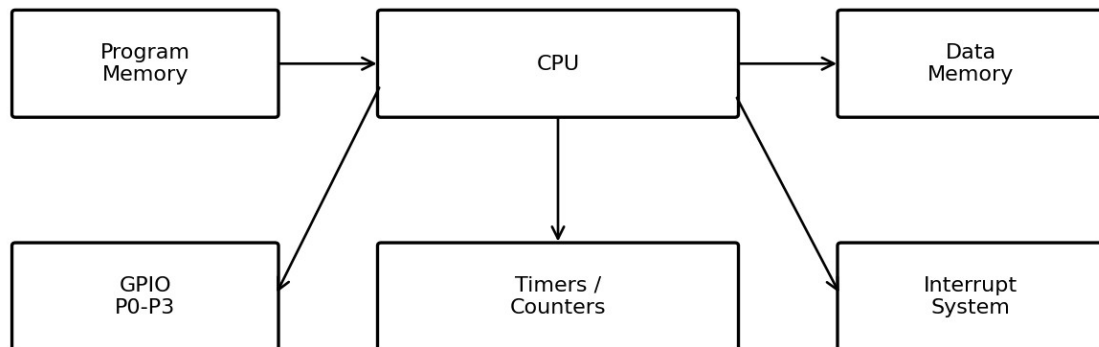
STC89C52 is a 8 bit microcontroller.

It is based on 8051 architecture.

It contains CPU, memory, registers, GPIO ports, timers and interrupts etc. It is used to control different electronic devices.

We use microcontroller because it contain many elements at once like CPU, memory, registers, I/O ports, timers etc. If it wasn't there, we should add every items separately and manually. Thats the main benefit of a microcontroller.

STC89C52 basic architecture



CPU Architecture

The CPU is the main part of the microcontroller. It is like the brain of the microcontroller. It receives instructions, understand them and perform the required task and produce the result.

The main parts of CPU are ALU, Control Unit, Registers, Program Counter, Stack Pointer and Status/Flag registers.

The CPU basically works in three main steps:

Fetch → Decode → Execute

Fetch means CPU get the instruction from the memory.

Decode means CPU understand what the instruction says.

Execute means CPU perform the operation.

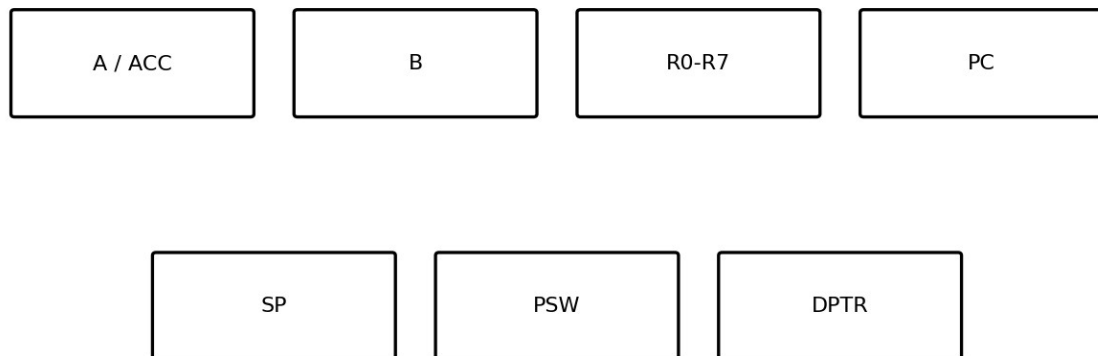
Registers

Registers are small storage locations inside the CPU. They are used for temporarily storing data and addresses during program execution.

The main registers are:

- Accumulator or A register
- B register
- R0 to R7 registers
- Program Counter
- Stack Pointer
- PSW

Main CPU registers



A Register / Accumulator

A register is a 8 bit register. It is also called as Accumulator or ACC.

It is mainly used for arithmetic, logical and data transfer operations. Many instructions use the A register for storing the result.

For example,

MOV A, #25H

Here 25H is loaded into the A register.

A register is mainly used in many operations and it is one of the important register in the CPU.

B Register

The B register is also a 8 bit register. It is mainly used along with A register for multiplication and division operations.

For multiplication:

MUL AB

For division:

DIV AB

A and B registers work together in these operations.

The B register can also be used as a general purpose register for some other operations.

R0 to R7 Registers

There are R0 to R7 registers. These are general purpose registers and each register is 8 bit.

They are used for temporarily storing data during program execution. They can also be used for counting and other small operations.

8051 architecture has multiple register banks. Each bank contains R0 to R7 and the active register bank is selected by PSW.

Program Counter (PC)

Next register is PC which means Program Counter.

PC is a 16 bit register. It stores the address of the next instruction which have to be fetched from the program memory.

When one instruction is executed, the PC normally move to the next instruction. But jump and branch instructions can change the PC value.

So simply we can say PC tells the CPU where the next instruction is located.

Stack Pointer (SP)

SP means Stack Pointer. It is a 8 bit register.

It is used to point the top of the stack. Stack is a memory area used for temporarily storing data.

The stack follows LIFO which means Last In First Out.

After reset, the SP value is 07H. When we use PUSH operation, SP is first increased and then the data is stored in the stack.

For example, if SP is 07H and we perform PUSH, SP becomes 08H and data is stored in that location.

SP is used during PUSH and POP operations. It is also involved when subroutines and interrupts need stack storage.

PSW / Program Status Word

PSW means Program Status Word. It is a special register which contains status information and control bits related to CPU operations.

It contains different flags like Carry, Auxiliary Carry and Overflow. It also contains register bank selection bits.

The important bits are:

- CY – Carry flag
- AC – Auxiliary Carry
- F0 – User flag
- RS1 and RS0 – Register bank selection
- OV – Overflow flag
- P – Parity flag

Carry Flag (CY)

Carry flag is used to show if a carry is generated from an arithmetic operation. For example when the result is more than the 8 bit limit, carry can be generated and CY becomes 1.

Auxiliary Carry (AC)

Auxiliary carry shows carry from the lower 4 bits to the upper 4 bits of an 8 bit operation. It is mainly useful in some arithmetic operations.

Overflow (OV)

Overflow flag is used to indicate overflow in signed arithmetic operations. When the result cannot be represented properly in the available signed range, overflow can occur.

Parity (P)

Parity flag is related with the number of 1 bits in the Accumulator. It is used to indicate the parity of the value present in A register.

RS1 and RS0

RS1 and RS0 are used to select the register bank. There are four register banks in 8051 architecture. Each bank contains R0 to R7 registers.

Memory

Memory is used for storing program and data.

There are mainly two types of memory:

1. Program Memory
2. Data Memory

Program Memory

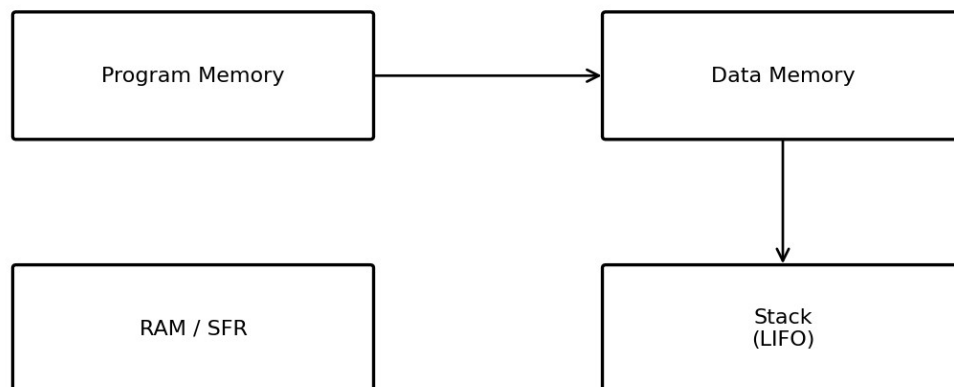
Program memory is used to store the instructions or program which CPU have to execute. CPU fetches the instructions from program memory using the Program Counter.

Data Memory

Data memory is used for storing data which is required during program execution. It can contain temporary data, registers and stack related storage.

The internal RAM contains register banks, bit addressable area and general purpose RAM.

Memory and stack idea



Stack

Stack is a special memory area used for temporarily storing data.

It follows LIFO - Last In First Out.

For example, if we PUSH A and then PUSH B, B will be taken first when POP operation is performed.

PUSH is used to put data into stack and POP is used to take data from stack.

Stack is also used during CALL, RET and interrupt operations.

SP is used to keep track of the current top of the stack.

Instruction Categories

Instruction set means the operations which CPU can perform.

The instructions can be divided into different categories.

1. Data Transfer

These instructions are used to move data from one place to another.

Examples: MOV, PUSH, POP

2. Arithmetic Instructions

These instructions are used to perform mathematical operations.

Examples: ADD, ADDC, SUBB, INC, DEC, MUL, DIV

3. Logical Instructions

These instructions are used for logical and bit related operations.

Examples: ANL, ORL, XRL, CLR, CPL

4. Branch / Control Instructions

These instructions are used to change the normal sequence of program execution.

Examples: SJMP, LJMP, AJMP, JZ, JNZ, DJNZ

5. Bit Operations

These instructions are used to operate on individual bits.

Examples: SETB, CLR, CPL

Basic Instruction Execution

When CPU gets an instruction, it process the instruction step by step.

The basic process is:

Fetch → Decode → Execute → Write Back

Fetch

CPU retrieve the next instruction from the program memory.

Decode

CPU understand the instruction and determine what operation it should perform.

Execute

CPU performs the operation given by the instruction.

Write Back

The result is stored in a register or memory.

For example:

MOV A, #25H

Before execution:

A = 00H

CPU fetch the instruction from program memory. Then it decode the instruction as a data transfer instruction. After that it execute the instruction and load 25H into A.

After execution:

A = 25H

GPIO

GPIO means General Purpose Input/Output.

GPIO pins are used for communicating with external devices.

A GPIO pin can work as:

Input → receive information from external device

Output → send signal to external device

STC89C52 has four main I/O ports:

- Port 0 → P0.0 to P0.7
- Port 1 → P1.0 to P1.7
- Port 2 → P2.0 to P2.7
- Port 3 → P3.0 to P3.7

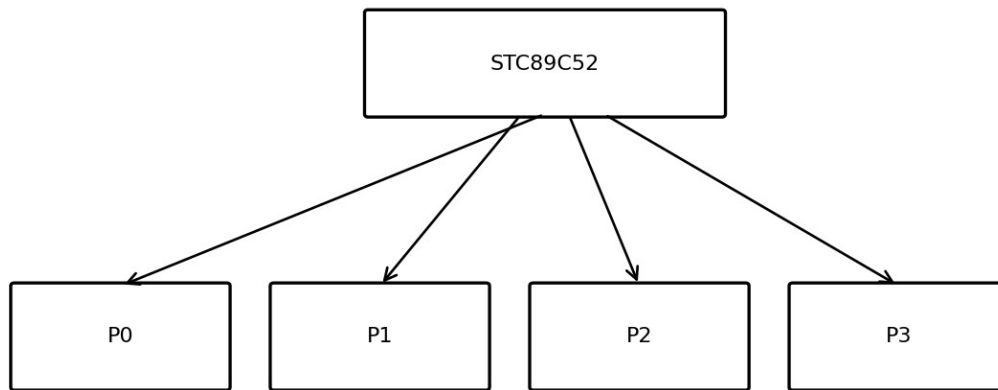
Each port has 8 pins.

So there are four 8 bit I/O ports.

For example, a button can be connected to a GPIO pin as input. When the button is pressed, the microcontroller can read the input.

An LED can be connected as output. The microcontroller can send high or low signal to control the LED.

GPIO ports



Timer / Counter

Timer is used for counting clock events inside the microcontroller.

Counter is used for counting external events.

Timers are useful for creating delays, timing operations and periodic tasks.

8051 architecture has Timer 0 and Timer 1.

Some important timer registers are:

TMOD

TCON

TH0 / TL0

TH1 / TL1

The timer can count the clock and when it reaches its limit, overflow can happen.

The timer overflow can also be used to generate an interrupt.

Interrupts

Interrupt is a mechanism which allows CPU to temporarily stop its current program and respond to an important event.

Normally CPU execute the main program continuously. If an interrupt occurs, CPU can stop the current work and execute the Interrupt Service Routine.

ISR means Interrupt Service Routine.

Simple flow is:

Main Program → Interrupt Occurs → Identify Source → Execute ISR → Return → Continue Main Program

Instead of continuously checking every device, an interrupt can tell the CPU that some event has happened.

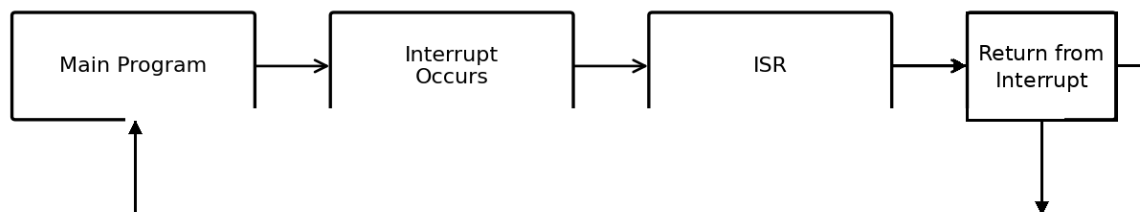
8051 compatible architecture provides interrupt sources such as:

- External Interrupt 0
- Timer 0
- External Interrupt 1
- Timer 1
- Serial communication

Interrupts can be enabled, disabled and prioritized using special registers like IE and IP.

After completing the interrupt service routine, CPU uses RETI and continue the main program.

Interrupt flow



Overall Working

The STC89C52 CPU gets instructions from program memory using the Program Counter.

The instruction is fetched, decoded and executed by the CPU.

Registers like A, B and R0-R7 are used for storing and processing data.

PSW stores the status and flag information of the CPU.

SP manages the stack which is used for temporary storage, subroutines and interrupts.

GPIO ports are used to communicate with external devices.

Timers are used for timing and counting operations.

Interrupts allow the CPU to respond to important events without continuously checking them.

So all these parts work together to make the STC89C52 microcontroller perform different tasks.